

Syntax

Column-Level

```
CREATE TABLE Employees(  
    EmployeeID INT PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL  
);
```

Most common syntax.

Table-Level

Unlike PRIMARY KEY or UNIQUE, NOT NULL cannot be declared as a table-level constraint in standard SQL.

It is always attached directly to a column.

2.3 Internal Working NULL Bitmap

Most database engines keep a small bitmap.

Imagine

Name	Age	Salary	Phone
0	1	0	1

could mean

NOT NULL	NULL	NOT NULL	NULL
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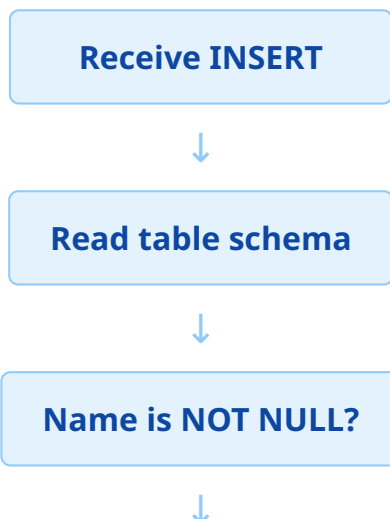
The actual implementation differs by database engine, but conceptually each row keeps metadata indicating which nullable columns contain NULL.

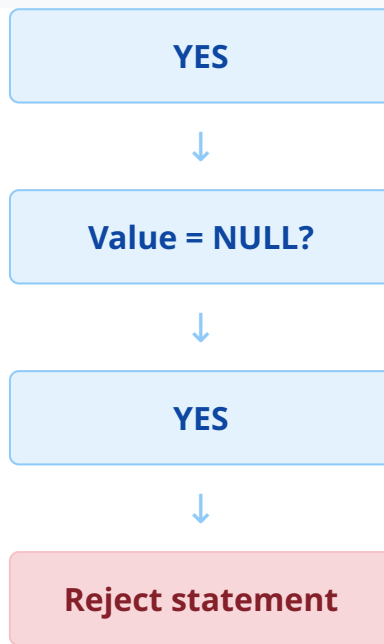
Validation Process

Suppose

```
INSERT INTO Employees(Name)VALUES(NULL);
```

Flow





Suppose

```
Name VARCHAR(100)
```

Later

```
ALTER TABLE Employees  
MODIFY Name VARCHAR(100) NOT NULL;
```

Existing NULL Values

Suppose

```
NULL  
NULL  
NULL
```

already exist.

Then

```
ALTER TABLE Employees
MODIFY Name VARCHAR(100) NOT NULL;
```

fails because existing rows violate the new constraint.

```
UPDATE Employees SET Name='Unknown'
WHERE Name IS NULL;
```

Then

```
ALTER TABLE Employees
MODIFY Name VARCHAR(100) NOT NULL;
```

very primary key is automatically NOT NULL

UNIQUE Constraint

3.1 Introduction

Purpose

Ensure that no two rows contain the same value in a column

PRIMARY KEY	UNIQUE
Only one per table	Many allowed
Never NULL	May allow NULLs (DBMS dependent)

Main identifier	Alternate identifier
Automatically NOT NULL	Nullable unless specified otherwise

Syntax

Single Column

```
CREATE TABLE Users(  
    Email VARCHAR(100) UNIQUE  
);
```

Multiple Columns

```
CREATE TABLE Employees(  
    CompanyID INT,  
    Email VARCHAR(100),  
  
    UNIQUE(CompanyID, Email)  
);
```

Each (CompanyID, Email) combination must be unique.

Named Constraint

```
CREATE TABLE Users(  
    Email VARCHAR(100),  
    CONSTRAINT uq_email UNIQUE(Email)  
);
```

Useful for maintenance and debugging.

NULL Handling

One of the most asked interview topics.

MySQL

Email UNIQUE

Allowed

NULL
NULL
NULL

Yes.

Multiple NULLs are allowed because NULL is considered "unknown," and unknown values are not treated as equal.

PostgreSQL

Same behavior.

Multiple NULLs allowed.

SQL Server

Different.

A traditional UNIQUE constraint allows only one NULL value.

Adding UNIQUE Later

```
ALTER TABLE Users  
ADD CONSTRAINT uq_email  
UNIQUE(Email);
```

If duplicates already exist, the statement fails until the duplicate data is cleaned up.

Removing UNIQUE

```
ALTER TABLE Users  
DROP INDEX uq_email;
```

(MySQL syntax.)